«interface»

IndexManager

+void initialize()
+void rebuildWeblogIndex()
+void rebuildWeblogIndex(Weblog website)
+void removeWeblogIndex(Weblog website)
+void addEntryIndexOperation(WeblogEntry entry)
+void addEntryReIndexOperation(WeblogEntry entry)
+void removeEntryIndexOperation(WeblogEntry entry)
+SearchResultList search(String term, String weblogHandle, String category, String locale, int pageNum, int entryCount, URLStrategy urlStrategy)
+boolean isInconsistentAtStartup()
+void release()
+void shutdown()

«interface»
Runnable

+void run()

LuceneIndexManager

- IndexReader reader
-Weblogger roller
- boolean searchEnabled
-String indexDir
- File indexConsistencyMarker
- boolean inconsistentAtStartup
- ReadWriteLock rwl
- Log logger

#LuceneIndexManager(Weblogger roller)
+void initialize()
+void rebuildWeblogIndex()
+void rebuildWeblogIndex(Weblog website)
+void removeWeblogIndex(Weblog website)
+void addEntryIndexOperation(WeblogEntry entry)
+void addEntryReIndexOperation(WeblogEntry entry)
+void removeEntryIndexOperation(WeblogEntry entry)
+SearchResultList search(String term, String weblogHandle, String category, String locale, int pageNum, int entryCount, URLStrategy urlStrategy)
+ReadWriteLock getReadWriteLock()
+boolean isInconsistentAtStartup()
+Analyzer getAnalyzer()
+void release()
+void shutdown()
+synchronized void resetSharedReader()
+synchronized IndexReader getSharedIndexReader()
+Directory getIndexDirectory()
-void scheduleIndexOperation(IndexOperation op)
-void executeIndexOperationNow(IndexOperation op)
- boolean indexExists()
-void deleteIndex()
-void createIndex(Directory dir)
-Analyzer instantiateAnalyzer()
-Analyzer instantiateDefaultAnalyzer()
-SearchResultList convertHitsToEntryList(ScoreDoc[] hits, SearchOperation search, int pageNum, int entryCount, String weblogHandle, boolean websiteSpecificSearch, URLStrategy urlStrategy)

IndexOperation

#Log logger
#LuceneIndexManager manager
- IndexWriter writer

+IndexOperation(LuceneIndexManager manager)
+void run()
#Document getDocument(WeblogEntry data)
#IndexWriter beginWriting()
#void endWriting()
#void doRun()

ReadFromIndexOperation

- Log logger

+ReadFromIndexOperation(LuceneIndexManager mgr)
+final void run()

WriteToIndexOperation

- Log logger

+WriteToIndexOperation(LuceneIndexManager mgr)
+void run()

AddEntryOperation

- Log logger
-WeblogEntry data
-Weblogger roller

+AddEntryOperation(Weblogger roller, LuceneIndexManager mgr, WeblogEntry data)
+void doRun()

ReIndexEntryOperation

- Log logger
-WeblogEntry data
-Weblogger roller

+ReIndexEntryOperation(Weblogger roller, LuceneIndexManager mgr, WeblogEntry data)
+void doRun()

RemoveEntryOperation

- Log logger
-WeblogEntry data
-Weblogger roller

+RemoveEntryOperation(Weblogger roller, LuceneIndexManager mgr, WeblogEntry data)
+void doRun()

RebuildWebsiteIndexOperation

- Log logger
-Weblog website
-Weblogger roller

+RebuildWebsiteIndexOperation(Weblogger roller, LuceneIndexManager mgr, Weblog website)
+void doRun()

RemoveWebsiteIndexOperation

- Log logger
-Weblog website
-Weblogger roller

+RemoveWebsiteIndexOperation(Weblogger roller, LuceneIndexManager mgr, Weblog website)
+void doRun()

SearchOperation

- Log logger
- String[] SEARCH_FIELDS
- Sort SORTER
- IndexSearcher searcher
-TopFieldDocs searchresults
-String term
- String weblogHandle
-String category
-String locale
- String parseError

+SearchOperation(IndexManager mgr)
+void setTerm(String term)
+void setWeblogHandle(String weblogHandle)
+void setCategory(String category)
+void setLocale(String locale)
+void doRun()
+IndexSearcher getSearcher()
+void setSearcher(IndexSearcher searcher)
+TopFieldDocs getResults()
+int getResultsCount()
+String getParseError()

IndexUtil

- IndexUtil()
+Term getTerm(String field, String input)

FieldConstants

+final String ANCHOR
+final String UPDATED
+final String ID
+final String USERNAME
+final String CATEGORY
+final String TITLE
+final String PUBLISHED
+final String CONTENT
+final String CONTENT_STORED
+final String C_CONTENT
+final String C_EMAIL
+final String C_NAME
+final String CONSTANT
+final String CONSTANT_V
+final String WEBSITE_HANDLE
+final String LOCALE

«external»

Weblogger

+WeblogEntryManager getWeblogEntryManager()
+WeblogManager getWeblogManager()
+ThreadManager getThreadManager()
+void release()

«external»

WeblogEntry

+String getId()
+String getTitle()
+String getText()
+Weblog getWebsite()
+User getCreator()
+WeblogCategory getCategory()
+List<WeblogEntryComment> getComments()
+Timestamp getPubTime()
+Timestamp getUpdateTime()
+String getLocale()

«external»

Weblog

+String getId()
+String getHandle()
+String getName()

«external»

WeblogEntryComment

+String getContent()
+String getEmail()
+String getName()

«external»

WeblogCategory

+String getName()

«external»
User

+String getUserName()

«external»

WeblogEntryManager

+WeblogEntry getWeblogEntry(String id)
+List<WeblogEntry> getWeblogEntries(WeblogEntrySearchCriteria criteria)

«external»

WeblogManager

+Weblog getWeblog(String id)

«external»

ThreadManager

+void executeInBackground(Runnable task)
+void executeInForeground(Runnable task)

«external»

URLStrategy

«external»
SearchResultList

+SearchResultList(List results, Set categories, int limit, int offset)

«external»

WeblogEntryWrapper

+WeblogEntryWrapper wrap(WeblogEntry entry, URLStrategy strategy)

«external»

WeblogEntrySearchCriteria

+void setWeblog(Weblog weblog)
+void setStatus(PubStatus status)

«external»

WebloggerConfig

+String getProperty(String key)
+boolean getBooleanProperty(String key, boolean defaultValue)
+int getIntProperty(String key)

«external»

WebloggerRuntimeConfig

+boolean isSiteWideWeblog(String handle)

«external»

WebloggerException

«external»

InitializationException

«external»
PubStatus

PUBLISHED

«external»
RollerConstants

+final long SEC_IN_MS

«Lucene»
IndexWriter

+void addDocument(Document doc)
+void deleteDocuments(Term term)
+void close()

«Lucene»
IndexReader

+DirectoryReader open(Directory dir)
+void close()

«Lucene»

DirectoryReader

+DirectoryReader open(Directory dir)
+boolean indexExists(Directory dir)

«Lucene»
IndexSearcher

+IndexSearcher(IndexReader reader)
+TopFieldDocs search(Query query, int n, Sort sort)
+Document doc(int docID)

«Lucene»
Document

+void add(Field field)
+Field getField(String name)

«Lucene»
Field

«Lucene»

StringField

+StringField(String name, String value, Store store)

«Lucene»
TextField

+TextField(String name, String value, Store store)

«Lucene»
SortedDocValuesField

+SortedDocValuesField(String name, BytesRef value)

«Lucene»
Term

+Term(String field, String text)

«Lucene»

Query

«Lucene»

TermQuery

+TermQuery(Term term)

«Lucene»

BooleanQuery

«Lucene»

BooleanQueryBuilder

+BooleanQuery.Builder add(Query query, BooleanClause.Occur occur)
+BooleanQuery build()

«Lucene»

MultiFieldQueryParser

+MultiFieldQueryParser(String[] fields, Analyzer analyzer)
+void setDefaultOperator(Operator op)
+Query parse(String query)

«Lucene»

Analyzer

+TokenStream tokenStream(String field, Reader reader)

«Lucene»

StandardAnalyzer

+StandardAnalyzer()

«Lucene»

LimitTokenCountAnalyzer

+LimitTokenCountAnalyzer(Analyzer delegate, int maxTokenCount)

«Lucene»
TokenStream

+void reset()
+boolean incrementToken()
+CharTermAttribute addAttribute(Class)

«Lucene»
CharTermAttribute

+String toString()

«Lucene»

Directory

+String[] listAll()

«Lucene»

FSDirectory

+FSDirectory open(Path path)
+String[] listAll()
+void close()

«Lucene»

IndexWriterConfig

+IndexWriterConfig(Analyzer analyzer)

«Lucene»

TopFieldDocs

+ScoreDoc[] scoreDocs
+TotalHits totalHits

«Lucene»
ScoreDoc

+int doc
+float score

«Lucene»
Sort

+Sort(SortField field)

«Lucene»
SortField

+SortField(String field, Type type, boolean reverse)

«Lucene»

BytesRef

+BytesRef(String text)

«Java»
ReadWriteLock

+Lock readLock()
+Lock writeLock()

«Java»
ReentrantReadWriteLock

+ReentrantReadWriteLock()
+Lock readLock()
+Lock writeLock()

«Java»
File

+boolean exists()
+boolean createNewFile()
+boolean delete()

«Java»
Path

+Path of(String path)

«Java»
Files

+void delete(Path path)

Singleton - Central coordinator for all
Lucene indexing and search operations.
Manages index lifecycle, consistency,
and thread-safe access.

Template Method Pattern:
- getDocument() is common logic
- doRun() is extension point
- run() orchestrates execution

Acquires READ lock before execution.
Allows concurrent read operations.
Never modifies index.

Acquires WRITE lock before execution.
Ensures exclusive access during writes.
Resets shared reader after completion.

Centralized field name constants.
Prevents magic strings.
Single source of truth.

Searches across CONTENT, TITLE, C_CONTENT fields.
Supports filtering by weblog, category, locale.
Results sorted by PUBLISHED date (descending).
Limited to 500 documents.

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rwl

1

indexConsistencyMarker

1

manager

1

creates/schedules

reader

0..1

roller

1

writer

0..1

uses

data

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roller

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data

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roller

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data

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roller

1

website

0..1

roller

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website

1

roller

1

searcher

0..1

searchresults uses

0..1

uses

website

1

1

creator

1

0..1

category

1

0..1

comments

1

0..*

manages

1

1

manages

1

1

manages

1

1

scoreDocs

1

0..*

creates creates creates creates creates creates

uses

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throws uses uses

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